

Arsh

Amherst, MA

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Education

University of Massachusetts Amherst

Manning College of Information and Computer Sciences

September 2025 – Present

Amherst, MA, U.S.A.

- Bachelor of Science in Computer Science
- Cumulative GPA: 3.963 / 4.00
- Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Software Development, Artificial Intelligence, Calculus II, Statistics, Reasoning Under Uncertainty

Scottish High International School

Indian School Certificate (ISC)

Graduated: May 2025

Gurugram, India

- Percentage Score: 94.6% – 4.0 / 4.0 GPA
- Relevant Coursework: Computer Science, Mathematics, Physics, Chemistry, English, Spanish, Hindi

Skills

Programming & Data: Python, SQL, Java, C, TypeScript, NumPy, Data Analysis, Data Visualization

Machine Learning & AI: TensorFlow, scikit-learn, AI Model Training, Neural Embeddings, Retrieval-Augmented Generation (RAG), Prompt Engineering

Generative AI: OpenAI API, Anthropic API, Google Gemini

Software & APIs: FastAPI, Flask, REST APIs, React, Angular, HTML, CSS

Developer Tools: Git, GitHub, VS Code, Claude Code, Codex, Copilot

Certifications: Google AI Essentials – Google | Claude 101 – Anthropic | Claude Code 101 – Anthropic

Work Experience

AI Market Research Intern

Aadrila technologies

June 2025 – August 2025

Mumbai, India

- Identified U.S. GenAI growth opportunities by applying quantitative market analysis, market sizing, competitive benchmarking, and customer segmentation with critical thinking and attention to detail.
- Accelerated research workflows and generated competitive intelligence by applying and evaluating AI models for market analysis with initiative, adaptability, and structured problem-solving.
- Supported go-to-market decisions by translating market and customer data into visual findings and recommendations through data analysis, data visualization, and concise communication.

Event Management Supervisor

SHISTECH

June 2023 – May 2025

Gurugram, India

- Led a 25+ member technical team across 2 editions, coordinating priorities and accountability to deliver successful events.
- Managed operations for 1,000+ participants, balancing priorities with organization, composure under pressure, and attention to detail.
- Resolved issues under tight deadlines through delegation, collaboration, conflict resolution, and time management.

Projects

FinSight | AI-Powered Financial Research Platform

- Engineered a full-stack React/FastAPI financial research platform indexing 17,986 SEC/BSE listing records with REST APIs, company search, interactive data visualization, and filing discovery.
- Implemented a local Retrieval-Augmented Generation (RAG) pipeline using neural embeddings, semantic similarity search, cosine distance, and Ollama to generate source-cited answers from five retrieved evidence passages per question.
- Built and executed 33 automated backend tests covering API validation, entity resolution, company matching, retrieval isolation, and citation integrity.

SignSprout | AI-Powered Hand Tracking Sign Language Tutor

- Developed an interactive ASL learning application with 71 lessons across 8 units, combining webcam-based computer vision, visual demonstrations, quizzes, and progress tracking for beginner sign language instruction.
- Implemented real-time gesture recognition and feedback for 26 letters, 10 numbers, and 24 vocabulary signs using TensorFlow.js, MediaPipe hand/body landmark tracking, pose estimation, and custom pose-and-motion matching.
- Improved model inference robustness through two-hand recovery, pose tolerance, and tracking-gap handling while executing webcam processing and privacy-preserving inference entirely on-device.